SMART INDIA HACKATHON 2026

Problem Statement ID : SIH26018

Problem Statement Title : Intelligent Land Record Digitization and Validation System

Theme : Smart Automation

PS Category : Software

Team ID : **SIH26-A0H-T309**

Team Name (Registered on portal) : SIXTH SENSE

One evidence-driven layer for citizens, government officers and future property discovery

EXTRACT → VALIDATE → RECONCILE → EXPLAIN → PRIORITISE → ACT

PROPOSED SOLUTION

Problem • prototype • differentiation • key capabilities

PROBLEM AT HAND

FRAGMENTED EVIDENCE
Deeds, RoR, mutation, registration, maps and legacy files can exist in different records/systems.

? HUMAN INTERPRETATION GAP
A citizen may possess documents but still cannot tell whether they agree or what a mismatch means.

G MANUAL OFFICER WORK
Staff must compare fields, dates and spatial evidence repeatedly before deciding which cases deserve attention.

M MARKET INFORMATION ASYMMETRY
Buyers depend on brokers, local claims or advertisements because property evidence is difficult to understand and compare.

OUR SOLUTION — WEB PROTOTYPE / DEMO

PROPERTY INTELLIGENCE / P-00421 — 92% READY

2.50 acres • Survey 142/3 • Coimbatore

RoR | SALE DEED | EC | MAP

FIELD	SOURCE VALUE	AI CHECK	MEANING
Owner	Ravi Kumar	MATCH	Identity aligned
Survey	142/3	MATCH	Parcel ID aligned
Area	2.50 vs 2.31 ac	REVIEW	7.6% difference
Timeline	2019→2023	MATCH	No break detected

AI EXPLANATION
“Survey area is 7.6% lower. This may indicate a measurement, subdivision/update or boundary-record issue. Verify the latest survey evidence.”

WHY WE STAND OUT / KEY FEATURES

01 Property Intelligence Profile
One structured profile links owner, parcel, documents, events and spatial evidence.

02 Evidence-linked AI
Every important finding points back to the fields/documents that produced it.

03 Conflict reasoning
Not just “mismatch”: what differs, where, how much, likely cause, severity and next check.

04 Citizen + Officer modes
Same evidence engine; simple explanation for citizens, investigation queue for officers.

05 Marketplace-ready trust
Verified attributes become the basis for buyer–property matching instead of only price/ads.

MARKETPLACE — SAME AI ENGINE, NEW DECISION

1. **SELLER** — Create property profile
2. **EVIDENCE** — Upload/link records
3. **VERIFY** — AI checks consistency
4. **LIST** — Publish verified attributes
5. **BUYER** — Describe needs in natural language
6. **MATCH** — AI ranks fit + evidence
7. **UNDERSTAND** — Buyer sees risks & reasons
8. **CONNECT** — Direct buyer ↔ seller

SIXTH SENSE

SEEING BEYOND RECORDS, SECURING EVERY LAND

TECHNICAL APPROACH

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Complete technical architecture and technology stack

DATA / EVIDENCE INPUTS

SCANNED RECORDS RoR • deeds • mutation • registration

HANDWRITTEN FILES legacy annotations • regional scripts

SPATIAL EVIDENCE cadastral map • parcel geometry • survey

USER INPUT seller/buyer details • purpose • budget

PROPERTY TRUTH ENGINE

1. **PREPROCESS** deskew • denoise • layout
2. **EXTRACT** OCR • CV • NLP • language
3. **NORMALIZE** names • dates • units • IDs
4. **LINK** property ↔ person ↔ document ↔ event
5. **RECONCILE** rules + cross-source comparison
6. **REASON** temporal + spatial + duplicate logic
7. **EXPLAIN** evidence + severity + next action

DECISION / USER OUTPUTS

CITIZEN MODE plain-language explanation • checklist • compare

OFFICER MODE risk queue • evidence graph • correction • audit

MARKETPLACE match • rank • risk explain • direct connect

ADMIN DASHBOARD accuracy • errors • pending review • district trends

FRONTEND

React / Next.js • responsive web UI • maps • dashboards • role-based screens

BACKEND

Python / FastAPI • validation APIs • inference orchestration • audit services

DATA

PostgreSQL • PostGIS • object storage • graph model • versioned evidence

AI / SECURITY

OCR + vision/LLM • confidence • RBAC • encryption • audit trail • human review

FEASIBILITY AND VIABILITY

Technical feasibility • implementation • safety • before / after

TECHNICAL FEASIBILITY

✓ OCR / Computer Vision / NLP / GIS are commercially available building blocks.
✓ Deterministic rules handle measurable checks; AI handles classification and explanation.
✓ Small labelled prototype can demonstrate the complete pipeline.
✓ Spatial comparison can begin with sample parcel polygons.

DATA ACCESS FEASIBILITY

PROTOTYPE → permitted/public/synthetic records for the demo.
PILOT → authorised departmental datasets / APIs.
PRODUCTION → connect to existing government LRMS, DILRMP, GIS, registration and other authorised databases through approved APIs/data-sharing.
The platform is an intelligence layer on top of official records not a replacement database.

ECONOMIC / OPERATIONAL VIABILITY

FACTOR	PROTOTYPE	SCALE-UP
Compute	Local / cloud	GPU / cloud
Data	Sample corpus	Authorised feeds
Review	Human queue	District workflow
Storage	Object store	Gov-grade infra
Revenue	Pilot / grant	SaaS / API / services

VALUE = fewer manual comparisons + faster exception review + clearer citizen decisions

IMPLEMENTATION VIABILITY

P1 sample corpus → P2 extraction + validation → P3 citizen/officer workflows → P4 marketplace → P5 authorised integration and district pilot.

MAINTENANCE / SAFETY

- Confidence threshold + human escalation.
- Source-linked evidence prevents unsupported claims.
- Versioning preserves historical records.
- RBAC separates citizen/officer/admin access.
- Audit trail logs AI output + human correction.
- AI result is decision support, not legal title certification.

BEFORE → AFTER

BEFORE	AFTER
Open multiple records	**One evidence profile**
Read manually	**AI extracts**
Compare fields	**Engine reconciles**
Find mismatch	**AI explains conflict**
Decide what to do	**Officer verifies exception**

IMPACT AND MARKET VALUE

Impact • India scale • government value • marketplace

SMART INDIA HACKATHON 2026

INDIA SCALE	6.30 LAKH	4.02 CRORE	4,148	DILRMP 3.0	Official DoLR dashboard / programme
	villages / CLR	digital maps	SROs computerised	2026–2031	

COMMON PEOPLE

- Understand property evidence in plain language.
- See exactly what disagrees before committing money.
- Reduce dependence on broker/local claims.
- Compare properties by trust + suitability, not only price.

GOVERNMENT / OFFICERS

- Exception-first review queue.
- Evidence-linked case reasoning.
- Less repetitive comparison and data entry.
- Analytics on recurring discrepancy patterns and record quality.

ECOSYSTEM / BUSINESS

- Banks: structured due diligence.
- Legal: property timeline/evidence organisation.
- Developers: acquisition screening.
- Government: intelligence/API layer for modernization.

MARKETPLACE — COMPLETE END-TO-END PROCESS

1. **SELLER ONBOARD** — Enter property + seller details
2. **EVIDENCE PACK** — Upload/link available records
3. **AI VERIFY** — Extract + reconcile + risk score
4. **TRUST PROFILE** — Verified fields + unresolved issues
5. **LIST PROPERTY** — Price + purpose + location + evidence status
6. **BUYER REQUEST** — “2,000 sq.ft under ₹40L near X”
7. **AI MATCH** — Fit score + evidence strength + risk
8. **BUYER EXPLAIN** — Why this property matched / what to verify
9. **DIRECT CONNECT** — Buyer ↔ seller; no broker required for discovery
10. **ESCALATE** — If high-risk → human/official verification path

SCALE / BUSINESS MODEL District pilot → State platform → National integrations | Gov/API licensing • institutional subscriptions • verification services • marketplace discovery

INTELLIGENT DIGITALIZATION

Deep dive: capture → preprocess → OCR/HTR → field intelligence → confidence → human review

01 CAPTURE
PDF / scan / photo

02 PREPROCESS
deskew • denoise
crop • contrast

03 LAYOUT AI
detect tables, regions, fields

04 OCR + HTR
printed + handwriting
multiple Indian languages

05 FIELD AI
owner • survey • area
mutation • registration

06 CONFIDENCE
field-level score
uncertain → review

INPUT PROBLEM — REAL RECORDS ARE MESSY

Example source:
- 1980s handwritten register
- faded ink + stamps + folds
- Tamil / regional terminology
- inconsistent column layouts
- abbreviations and overwritten values
- same person may appear with spelling variations

A normal scanner only creates another image. Our system creates usable evidence.

WHAT OUR AI ACTUALLY DOES

1. Image quality assessment → repair only when useful.
2. Layout model identifies title, table, row and field regions.
3. OCR/handwriting model reads text in the detected regions.
4. NLP maps raw text to a standard land schema.
5. Entity resolution links spelling variants to candidate identities.
6. Unit/date/ID normalization creates comparable values.
7. Each field receives a confidence score + source location.
8. Low-confidence values enter human verification.

BETTER THAN ORDINARY DIGITIZATION

ORDINARY OCR
Image → text → database
Risk: errors become invisible structured data.

OUR PIPELINE
Document → structured field → confidence → evidence link → validation → explanation

Key difference:
We preserve the connection between a digital value and the exact source evidence that produced it.

Prototype output: side-by-side source document + extracted form + highlighted uncertain fields.

EXAMPLE "2.50 ACRES" → FIELD VALUE 2.50 → UNIT ACRES → SOURCE BOX [page 3, row 18] → CONFIDENCE 96% → READY FOR VALIDATION

SIXTH SENSE

SEEING BEYOND RECORDS, SECURING EVERY LAND

VERIFICATION AND VALIDATION

Deep dive: rules → cross-record → temporal → spatial → duplicate reasoning → action

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SIH

PROPERTY P-00421=> OWNER Ravi Kumar SURVEY 142/3 AREA 2.50 acres STATUS REVIEW

BUSINESS RULES	Area > 0 • valid survey format • dates logical • mandatory fields
CROSS-RECORD	Owner / survey / area / classification / mutation agreement
TEMPORAL	Sale → registration → mutation → current record chronology
SPATIAL	Recorded area vs mapped area • overlap • parcel geometry
DUPLICATE / ENTITY	Duplicate survey IDs • repeated owners • possible duplicate records

?

FROM MISMATCH TO REASON

OBSERVED
Record = 2.50 acres
Map = 2.31 acres
Difference = 0.19 acres
≈ 7.6% lower

POSSIBLE EXPLANATIONS
- outdated geometry
- subdivision/update
- boundary measurement difference
- extraction error

AI must not choose a legal conclusion without evidence.

✓

ACTIONABLE OUTPUT

SEVERITY
REVIEW — material but explainable candidate

EVIDENCE
→ record page + extracted field
→ map geometry
→ relevant transaction/mutation event

NEXT CHECK
"Verify the latest survey / subdivision evidence before relying on the area."

OFFICER QUEUE
High-impact + low-confidence cases rise to the top.

AUDIT
AI finding + source + human decision are stored together.

VALIDATION ENGINE = CHECKS RULES + CROSS-SOURCE COMPARISON + SPATIAL/TEMPORAL REASONING + GIVES CONFIDENCE RATING

AI SUPPORT ≠ LEGAL CERTIFICATION

MARKETPLACE

Deep dive: seller → evidence → verification → trust profile → AI matching → direct discovery

S

SELLER / PROPERTY SIDE

1	ONBOARD	Seller enters property basics location • size • price • purpose
2	EVIDENCE PACK	Upload available documents and spatial evidence
3	AI VERIFY	Extract → reconcile → confidence → unresolved issues
4	TRUST PROFILE	Verified attributes + risk flags + evidence strength
5	LIST	Property becomes searchable with transparent evidence status

B

BUYER / REQUIREMENT SIDE

A	DESCRIBE NEED	“2,000 sq.ft near X under ₹40 lakh for home”
B	NORMALISE	AI converts natural language into structured preferences
C	MATCH	Location + size + budget + purpose + evidence quality
D	RANK	Fit score + risk + evidence strength —not just advertisement ranking
E	EXPLAIN	“Why this matched” + what buyer still needs to verify

MATCH

AFTER MATCH

DIRECT CONNECT → buyer and seller communicate → buyer sees evidence/risk explanation → high-risk cases can be escalated for official/human verification → transaction-stage checks preserve the evidence trail

UNIQUE VALUE: VERIFIED ATTRIBUTES BECOME THE TRUST SIGNAL OF THE MARKETPLACE — THE SAME DATA SERVES GOVERNMENT WORKFLOW AND CITIZEN DISCOVERY.

RESEARCH AND REFERENCES

Official problem basis • government ecosystem • technical foundation • research links

INDIA LAND-RECORD ECOSYSTEM

DILRMP 3.0
2026–2031

6.30 LAKH
villages / CLR

4.02 CRORE
digital maps

4,148
SROs computerised

RESEARCH GAP

Digitisation alone stores records. Sixth Sense adds an evidence-linked intelligence layer that reconciles conflicts, explains them and routes action.

REFERENCES

01	**SIH 2026 PS SIH26018**	Intelligent Land Record Digitization and Validation System
02	**Department of Land Resources**	DILRMP — objectives, components, implementation
03	**DILRMP 3.0 Operational Guidelines**	2026–2031 programme phase
04	**DoLR Current Dashboard**	CLR / digital maps / SRO computerisation indicators
05	**Bhoomi Samman**	RoR, cadastral maps, registration and integration components

TECHNICAL BASIS / EVALUATION PLAN

OCR / DOCUMENT AI	Printed + handwritten recognition; layout-aware field extraction.
COMPUTER VISION	Map/image interpretation, seals/layout features and visual evidence.
NLP / LLM	Field classification + evidence-grounded natural-language explanation.
POSTGIS / GIS	Parcel geometry, overlap, boundary and area comparison.
KNOWLEDGE GRAPH	Property ↔ person ↔ document ↔ transaction/event relationships.
HUMAN-IN-LOOP	Confidence thresholds route uncertain/high-impact cases for review.
EVALUATION	Field accuracy • conflict precision/recall • calibration • explanation correctness • review-time reduction.

LINKS

SIH26018   DoLR / DILRMP   DILRMP 3.0 Guidelines

[SIH26018]   [DoLR / DILRMP]   [DILRMP 3.0]